

DLR-TomoSAR — Loss theory

two invariances · the joint error · the recall paradox · the composition ambiguity · L1 vs MSE

July 2026

One chain — from the label to the channels, and back

Take one pixel and follow its parameters through the physics. Every symbol used in this discussion is introduced on its own line:

$$p(z) = \sum_k a_k \mathcal{N}(z; \mu_k, \sigma_k)$$

the label: scales a_k , elevations μ_k , widths σ_k , along elevation z

$$\implies \mathbb{E}[\gamma(z) \gamma^*(z')] = p(z) \delta(z-z')$$

γ : the pixel's speckled reflectivity — random, with power profile p

$$\implies g_n = \int \gamma(z) e^{jk_n z} dz = A_n e^{j\theta_n}$$

pass n : one Fourier coefficient, at frequency $k_n \propto$ baseline $b_{\perp, n}$

$$\implies \underbrace{A_0, \dots, A_4}_{5 \text{ amplitudes}}, \quad \underbrace{\varphi_n = \theta_0 - \theta_n}_{4 \text{ IFG phases}}$$

the nine input channels: $A_n = |g_n|$, $\theta_n = \arg g_n$

$$\implies \hat{R}_{mn} = \langle A_m A_n e^{j(\varphi_n - \varphi_m)} \rangle_{B(x)} \approx P \rho(k_m - k_n)$$

covariance over the window $B(x)$: θ_0 cancels; $P = \int p$, $\rho = \text{FT}[p/P]$

$$\implies \text{Capon}(\hat{R}) \rightarrow \hat{p} \rightarrow \text{fit} \rightarrow a_k, \mu_k, \sigma_k$$

the GT pipeline — the same parameters return

The loop closes: the nine channels contain everything the label was built from. An ablation can only *delete* information — the next two chains show which parameter each half can never see.

Two exact invariances

- **Amplitudes are shift-blind.** Move the whole profile up by Δ :

$$p(z) \rightarrow p(z - \Delta)$$

$$\implies \gamma(z) \rightarrow \gamma(z - \Delta)$$

$$\implies g_n \rightarrow e^{jk_n\Delta} g_n$$

$$\implies A_n = |e^{jk_n\Delta}| |g_n| \text{ unchanged}$$

Position never reaches an amplitude — 29 of them are as blind as 5. What they still carry:

$$E[A_n^2] = P \implies \text{the scale } a$$

$$\text{cov}(A_m^2, A_n^2) = P^2 |\rho(k_m - k_n)|^2 \implies \sigma, \Delta\mu$$

Scale and shape (σ , separation $\Delta\mu$) survive; the position μ is gone.

- **Phases are scale-blind.** Scale the whole profile by $c > 0$:

$$p(z) \rightarrow c p(z)$$

$$\implies \gamma(z) \rightarrow \sqrt{c} \gamma(z)$$

$$\implies g_n \rightarrow \sqrt{c} g_n$$

$$\implies \varphi_n = \theta_0 - \theta_n \text{ unchanged}$$

Scale never reaches a phase. What they still carry — the shape ρ at frequency κ , for one scatterer at μ , width σ :

$$\rho(\kappa) = e^{j\kappa\mu} e^{-\kappa^2\sigma^2/2}$$

$$\implies \arg \rho(k_n) = k_n \mu \implies \text{a line of slope } \mu$$

$$\implies |\rho(k_n)| = e^{-k_n^2\sigma^2/2} \implies \text{phase spread} \rightarrow \sigma$$

Position is read directly; a second scatterer bends the line — detection.

Putting the two invariances together

- **The shift error.** One component $\mathcal{N}_\mu = \mathcal{N}(z; \mu, \sigma)$ of energy $E = \|\mathcal{N}_\mu\|^2$; the prediction lands δ away:

$$\|\mathcal{N}_{\mu+\delta} - \mathcal{N}_\mu\|^2 = 2E - 2\langle \mathcal{N}_{\mu+\delta}, \mathcal{N}_\mu \rangle$$

$$\Rightarrow \langle \mathcal{N}_{\mu+\delta}, \mathcal{N}_\mu \rangle = E e^{-\delta^2/4\sigma^2}$$

$$\Rightarrow \|\mathcal{N}_{\mu+\delta} - \mathcal{N}_\mu\|^2 = 2E(1 - e^{-\delta^2/4\sigma^2})$$

$$\Rightarrow \delta \ll \sigma : \approx E\delta^2/2\sigma^2, \quad \delta \gg \sigma : \rightarrow 2E$$

$\langle f, g \rangle = \int fg$: the overlap; $\|f\|^2 = \langle f, f \rangle$: the energy — only the cross-term moves.

- **The scale error.** The same component, its brightness off by the relative error ε :

$$\|(1+\varepsilon)\mathcal{N}_\mu - \mathcal{N}_\mu\|^2 = \|\varepsilon\mathcal{N}_\mu\|^2 = \varepsilon^2 E$$

$$\Rightarrow \varepsilon^2 E = 2E \text{ only at } \varepsilon = \sqrt{2} \text{ — a 141\% error}$$

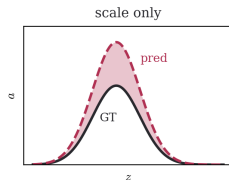
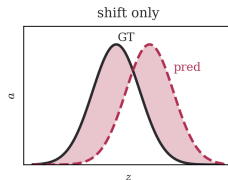
- **Take the derivative of each error:**

$$\frac{d}{d\delta} \|\mathcal{N}_{\mu+\delta} - \mathcal{N}_\mu\|^2 = \frac{E\delta}{\sigma^2} e^{-\delta^2/4\sigma^2} \xrightarrow{\delta \gg \sigma} 0$$

$\Rightarrow$ the shift error flattens: $2E$ is a true maximum

$$\frac{d}{d\varepsilon} \|(1+\varepsilon)\mathcal{N}_\mu - \mathcal{N}_\mu\|^2 = 2\varepsilon E > 0 \text{ at every } \varepsilon$$

$\Rightarrow$ the scale error never flattens



Shift and scale together — the joint error

- Both mistakes at once — the prediction scaled by $1+\varepsilon$ and misplaced by δ :

$$\|(1+\varepsilon)\mathcal{N}_{\mu+\delta} - \mathcal{N}_{\mu}\|^2$$

$$\Rightarrow = (1+\varepsilon)^2 E + E - 2(1+\varepsilon) \langle \mathcal{N}_{\mu+\delta}, \mathcal{N}_{\mu} \rangle$$

$$\Rightarrow = E[(1+\varepsilon)^2 + 1 - 2(1+\varepsilon)e^{-\delta^2/4\sigma^2}]$$

$$\Rightarrow \varepsilon=0: 2E(1 - e^{-\delta^2/4\sigma^2}), \quad \delta=0: \varepsilon^2 E$$

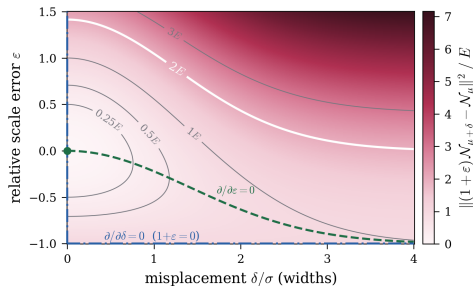
- Take the partial derivatives:

$$\frac{\partial}{\partial \delta} \|(1+\varepsilon)\mathcal{N}_{\mu+\delta} - \mathcal{N}_{\mu}\|^2 = (1+\varepsilon) \frac{E\delta}{\sigma^2} e^{-\delta^2/4\sigma^2}$$

$$\frac{\partial}{\partial \varepsilon} \|(1+\varepsilon)\mathcal{N}_{\mu+\delta} - \mathcal{N}_{\mu}\|^2 = 2E(1+\varepsilon - e^{-\delta^2/4\sigma^2})$$

$$\Rightarrow \frac{\partial}{\partial \delta} = 0 \text{ at } \delta = 0 \text{ (any } \varepsilon)$$

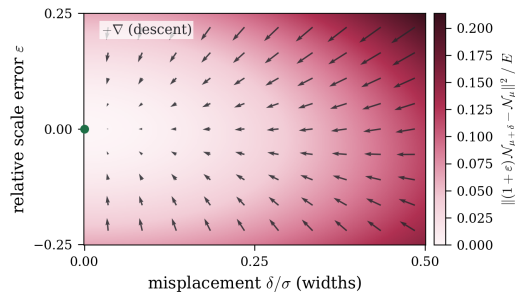
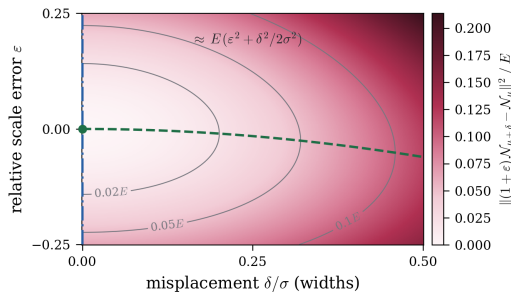
$$\Rightarrow \frac{\partial}{\partial \varepsilon} = 0 \text{ at } 1+\varepsilon = e^{-\delta^2/4\sigma^2} < 1$$



the joint landscape: the critical curves meet only at the truth — the surface's one minimum.

The bowl and the training pull

Zoom into small δ, ε — the regime of a converging network:



left: the quadratic bowl — elliptic contours, valley tangent to $\varepsilon=0$, no plateau. right: the $-\nabla$ descent field — onto the valley at curvature $2E$, along it at E ; its dim drift feeds the recall paradox (two frames ahead).

The full profile — separated scatterers decouple

- **Write the error as residuals.** Truth and prediction are sums of K components; Hungarian pairing matches them one to one, so pair k owns its shift δ_k and scale error ε_k :

$$p = \sum_k \mathcal{N}_{\mu_k}, \quad \hat{p} = \sum_k (1 + \varepsilon_k) \mathcal{N}_{\mu_k + \delta_k}$$

$$\implies \hat{p} - p = \sum_k r_k, \quad r_k = (1 + \varepsilon_k) \mathcal{N}_{\mu_k + \delta_k} - \mathcal{N}_{\mu_k}$$

$\mathcal{N}_{\mu_k}$: the k -th component $a_k \mathcal{N}(z; \mu_k, \sigma_k)$, energy

$E_k = \|\mathcal{N}_{\mu_k}\|^2 = a_k^2 / (2\sqrt{\pi} \sigma_k)$; r_k : its residual.

- **Square the sum** — bilinearity, no approximation:

$$\|\hat{p} - p\|^2 = \sum_k \|r_k\|^2 + 2 \sum_{j < k} \langle r_j, r_k \rangle$$

$$\|r_k\|^2 = E_k [(1 + \varepsilon_k)^2 + 1 - 2(1 + \varepsilon_k) e^{-\delta_k^2 / 4\sigma_k^2}]$$

- **Assume the scatterers are separated.** Any two Gaussian components overlap in closed form — exponentially small in the distance between their centres m, m' :

$$\langle a_j \mathcal{N}(z; m, \sigma_j), a_k \mathcal{N}(z; m', \sigma_k) \rangle$$

$$\propto e^{-(m - m')^2 / 2(\sigma_j^2 + \sigma_k^2)}$$

$\langle r_j, r_k \rangle$ is four such overlaps, centred near μ_j and μ_k .

Assume the separation is big enough — a few joint widths — and all four cancel:

$$|\mu_j - \mu_k| \geq 3 \sqrt{\sigma_j^2 + \sigma_k^2}$$

$$\implies e^{-(\mu_j - \mu_k)^2 / 2(\sigma_j^2 + \sigma_k^2)} \leq e^{-9/2} \approx 0.01$$

$$\implies \langle r_j, r_k \rangle \approx 0 \implies \|\hat{p} - p\|^2 \approx \sum_k \|r_k\|^2$$

The recall paradox — MSE turns doubt into dimming

- **Squared error is minimised by the conditional mean** — the value the loss pulls every output toward:

$$\hat{a}_2(x) = E[a_2 | x]$$

x : the input patch; a_2 : the second amplitude label — defined as 0 on pixels with a single scatterer ($K=1$).

- **Split the mean over the two cases.** The label is zero in one case and bright in the other; the zero case drops:

$$\begin{aligned}\hat{a}_2 &= \Pr(K=2 | x) E[a_2 | K=2, x] \\ &\quad + \Pr(K=1 | x) E[a_2 | K=1, x] \\ &= \Pr(K=2 | x) \cdot E[a_2 | K=2, x]\end{aligned}$$

- **Bayes' rule**, with $\pi = \Pr(K=2)$, and total probability over the two cases:

$$\begin{aligned}\Pr(K=2 | x) &= \frac{p(x | K=2) \pi}{p(x | K=2) \pi + p(x | K=1) (1-\pi)} \\ &= \frac{\pi \Lambda}{1-\pi + \pi \Lambda}, \quad \Lambda(x) = \frac{p(x | K=2)}{p(x | K=1)}\end{aligned}$$

- **$|A|$ runs:**

$$\Lambda(x) \approx 1 \text{ on every pixel} \Rightarrow \Pr(K=2 | x) \approx \pi$$

$$\Rightarrow \hat{a}_2 \text{ pulled to } \pi \cdot E[a_2 | K=2, x]$$

$$\Rightarrow E[a_2 | K=2, x] = f_\theta(P(x))$$

$$\hat{a}_2 = \pi \cdot f_\theta(P(x)) \text{ — dimmed}$$

$$\Rightarrow f_\theta(P(x)) \rightarrow 0, \text{ class imbalance}$$

- **φ -only:** every pixel lands at an extreme:

$$\Lambda(x) \gg 1 \text{ on true pairs} \Rightarrow \Pr(K=2 | x) \rightarrow 1$$

$$\Rightarrow \hat{a}_2 \rightarrow E[a_2 | K=2, x] \text{ — undimmed}$$

$$\hat{a}_2 = E[a_2 | K=2, x] = E[a_2 | K=2] = \bar{a}_2^{\text{active}}$$

$$\Lambda(x) \ll 1 \text{ on singles} \Rightarrow \Pr(K=2 | x) \rightarrow 0$$

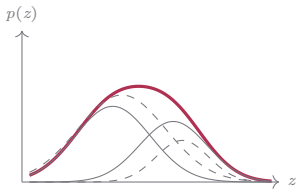
$$\Rightarrow \hat{a}_2 \rightarrow 0 \text{ — the slot closes}$$

Why parameter losses win: the composition ambiguity

A curve loss supervises only the **composition** $\sum_k a_k g_k(z)$ — the model never sees the components it must actually emit.

- Many different parameter changes produce the same curve change: widen one Gaussian, or raise its neighbour; shift μ , or trade amplitudes.
- The gradient distributes credit *ambiguously* across slots — the composition is many-to-one in the parameters.
- A parameter loss (after Hungarian pairing) supervises each Gaussian **directly**: unambiguous per-slot credit assignment.

The curve is what we care about — but the better route to a good curve is supervising its constituents.



Two different component sets (solid vs dashed) compose nearly the same profile (thick): a curve loss can barely tell them apart.

Two ways to route the error signal

curve-space loss — one residual, broadcast to every slot

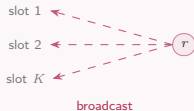
differentiate — the loss $\ell_{\text{curve}} = \sum_n (\hat{p}(\xi_n) - p^{\text{gt}}(\xi_n))^2$ sees only the composition; one shared residual factor:

$$\frac{\partial \ell_{\text{curve}}}{\partial \hat{\theta}_k} = 2 \sum_n \underbrace{r(\xi_n)}_{\text{same for every } k} \frac{\partial \hat{p}(\xi_n)}{\partial \hat{\theta}_k}, \quad r = \hat{p} - p^{\text{gt}}$$

project — each derivative is the slot's own kernel g_k times a low-order polynomial q_θ :

$$\begin{aligned} \frac{\partial \hat{p}}{\partial \hat{a}_k} &= g_k, \\ \frac{\partial \hat{p}}{\partial \hat{\mu}_k} &= \hat{a}_k g_k \frac{\xi - \hat{\mu}_k}{\hat{\sigma}_k^2}, \\ \frac{\partial \hat{p}}{\partial \hat{\sigma}_k} &= \hat{a}_k g_k \frac{(\xi - \hat{\mu}_k)^2}{\hat{\sigma}_k^3} \end{aligned}$$

$$\implies \frac{\partial \ell_{\text{curve}}}{\partial \hat{\theta}_k} = 2 \langle r, q_\theta g_k \rangle \quad \text{— no slot identity in the error signal}$$



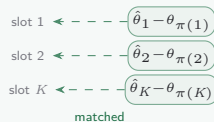
parameter-space loss — K private errors

match, then differentiate — π supplies identity:

$$\ell_{\text{param}} = \sum_k w_k |\hat{\theta}_{\pi(k)} - \theta_k|$$

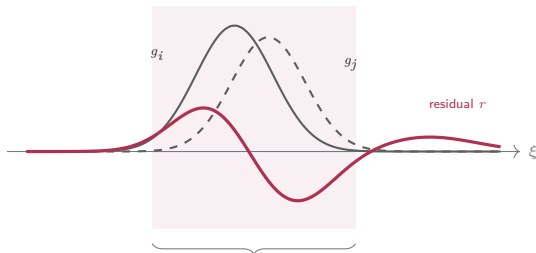
$$\frac{\partial \ell_{\text{param}}}{\partial \hat{\theta}_{\pi(k)}} = w_k \text{sign}(\hat{\theta}_{\pi(k)} - \theta_k)$$

the Jacobian is **diagonal**: each Gaussian corrects its *own* error — no shared factor, no credit dispute.



The composition is many-to-one in the parameters: the broadcast cannot name the slot that must fix the error — the matching can.

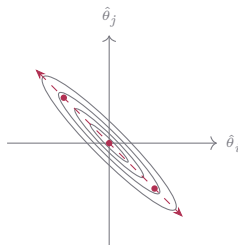
Overlap degenerates the broadcast — the matched loss stays diagonal



overlapping kernels read r through the same window

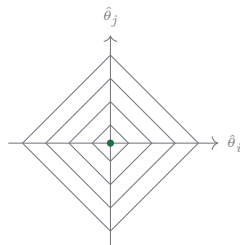
$$\partial \ell / \partial \hat{\theta}_i = 2 \langle r, q g_i \rangle \approx 2 \langle r, q g_j \rangle = \partial \ell / \partial \hat{\theta}_j$$

— the two slots' gradients become **collinear**.



curve loss near overlap: a **flat valley**

— every dot fits equally well



param-L1: **one minimum**

— each slot pulled to its target

The grid made this mechanism measurable: on the grid average every parameter-space objective beats every curve-space objective; the set-pred head is what lets the robust curve losses recover.

Does the ambiguity argument survive set prediction?

How much does moving from a curve loss to a parameter loss buy, per head?

head	advantage of param over curve losses (composite)
sorted conv regression	+0.13 — all 11 backbones agree
Hungarian set prediction	+0.02 — within noise, all 3 backbones

One cell makes the mechanism visible — same backbone, same loss, only the head changes (leaderboard ranks, out of 124):

ResUNet + Huber-curve	curve fit	component shape	peak location
sorted head	3rd	102nd	102nd
set-pred head	1st	52nd	29th

The argument stands, with a scope: it describes the sorted column of the grid. Set prediction changes who supplies slot identity, not whether identity is needed.

How set prediction closes the gap: the gradient view

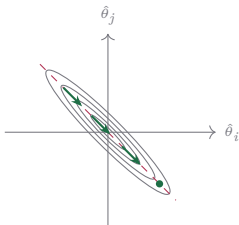
$$\hat{p}(\xi) = \sum \sigma(\hat{e}_k) \hat{a}_k g_k(\xi) \quad \text{gated composition}$$

$$\ell = \ell_{\text{curve}}(\hat{p}) + \lambda \sum_k \text{BCE}(\sigma(\hat{e}_{\pi(k)}), e_k), \quad \pi = \text{assign}(\hat{\theta}, \theta)$$

$$\frac{\partial \ell}{\partial \hat{e}_{\pi(k)}} = \underbrace{2 \langle r, q_e g_{\pi(k)} \rangle}_{\text{broadcast}} + \underbrace{\lambda (\sigma(\hat{e}_{\pi(k)}) - e_k)}_{\text{diagonal}}$$

$\sigma(\cdot)$ = logistic gate (distinct from the width $\hat{\sigma}_k$); $q_e = \sigma'(\hat{e}) \hat{a}$ — the gating path;

$\partial \ell / \partial \hat{\theta}_k = 2 \langle r, q_{\theta} g_k \rangle$ stays a broadcast, exactly as before.



The gated head adds **one supervised coordinate per slot**: an existence gate $\sigma(\hat{e}_k)$, assigned to its target e_k by a slot assignment π (sorted or Hungarian — identical at $K=2$).

- The curve term is unchanged — its error signal still carries no slot identity.
- The presence term is **diagonal**: one private error per slot — the same structure the param loss enjoys, restricted to one coordinate.
- That coordinate breaks the tie. The degenerate points on the curve valley differ in *cardinality* (a split occupies an extra active slot), so the presence gradient is nonzero exactly along the flat direction: the valley tilts toward the correct-cardinality solution (figure).
- π is recomputed from prediction–target proximity every step, so the private errors follow the slots. A sorted assignment is wired by GT rank instead: when slots drift, it pulls them toward the wrong targets (the $K=3$ instability).

Why param-L1 beats param-MSE — the label laws this pipeline manufactures

The param loss runs after the matcher, in normalised parameter space, under this head's supervision rule:

$$\ell_{\text{param}} = \sum_k \sum_{p \in \{a, \mu, \sigma\}} m_k^{(p)} \ell(\hat{\theta}_{\pi(k)}^{(p)}, \theta_k^{(p)})$$
$$m_k^{(a)} = 1 \text{ always,} \quad m_k^{(\mu)} = m_k^{(\sigma)} = \mathbf{1}[a_k^{\text{gt}} > 10^{-3}]$$

- **The amplitude law is spike-and-slab by construction.**

$a = 0$ (empty slot), $a > 0$ (occupied)

$$\begin{aligned} \implies p(a | x) &= \Pr(a=0 | x) p(a | a=0, x) \\ &\quad + \Pr(a>0 | x) p(a | a>0, x) \end{aligned}$$

$$\implies p(a | a=0, x) = \delta(a), \quad p(a | a>0, x) \text{ a density}$$

$$\begin{aligned} \implies p(a | x) &= \Pr(a=0 | x) \delta(a) \\ &\quad + \Pr(a>0 | x) p(a | a>0, x) \end{aligned}$$

- **The position law is two-valued where layover strikes.**

two-scatterer cell: $\mu \in \{\mu_1, \mu_2\}, \quad \mu_1 < \mu_2$

$$\begin{aligned} \implies p(\mu | x) &= \Pr(\mu=\mu_1 | x) \delta(\mu-\mu_1) \\ &\quad + \Pr(\mu=\mu_2 | x) \delta(\mu-\mu_2) \end{aligned}$$

$$\implies \mu_2 - \mu_1 \text{ — set by the scene, not the model}$$

Why param-L1 beats param-MSE — MSE trains the conditional mean

- **One instrument from decision theory:** training drives the output c to the minimiser of the conditional risk $R(c | x) = \mathbb{E}[\ell(c, \theta) | x] = \int \ell(c, \theta) p(\theta | x) d\theta$. Evaluate it for the squared loss, one manipulation per line:

$$\ell(c, \theta) = (c - \theta)^2$$

the loss

$$\implies R(c | x) = \mathbb{E}[(c - \theta)^2 | x]$$

plug ℓ into the risk

$$\implies (c - \theta)^2 = c^2 - 2c\theta + \theta^2$$

expand the square

$$\implies R(c | x) = \mathbb{E}[c^2 - 2c\theta + \theta^2 | x]$$

inside the expectation

$$\implies R(c | x) = c^2 - 2c\mathbb{E}[\theta | x] + \mathbb{E}[\theta^2 | x]$$

linearity of $\mathbb{E}$; c is a constant given x

$$\implies \frac{dR}{dc} = 2c - 2\mathbb{E}[\theta | x]$$

differentiate in c ; $\mathbb{E}[\theta^2 | x]$ is constant

$$\implies \frac{dR}{dc} = 0 \iff c^* = \mathbb{E}[\theta | x]$$

set the slope to zero: the conditional mean

Why param-L1 beats param-MSE — L1 trains the conditional median

- **The same conditional risk**, evaluated for the absolute loss:

$$\ell(c, \theta) = |c - \theta|$$

the loss

$$\implies R(c | x) = \int |c - \theta| p(\theta | x) d\theta$$

plug ℓ into the risk

$$\implies |c - \theta| = c - \theta \quad (\theta < c), \quad \theta - c \quad (\theta > c)$$

resolve the absolute value

$$\implies R(c | x) = \int_{\theta < c} (c - \theta) p d\theta + \int_{\theta > c} (\theta - c) p d\theta$$

split the integral at c

$$\implies \frac{d}{dc} \int_{\theta < c} (c - \theta) p d\theta = \int_{\theta < c} p d\theta + (c - c) p(c)$$

Leibniz: integrand, plus the moving boundary

$$\implies \frac{d}{dc} \int_{\theta > c} (\theta - c) p d\theta = - \int_{\theta > c} p d\theta - (c - c) p(c)$$

the same on the upper piece

$$\implies (c - c) p(c) = 0$$

both boundary terms vanish

$$\implies \frac{dR}{dc} = \Pr(\theta < c | x) - \Pr(\theta > c | x)$$

the two masses remain

$$\implies \frac{dR}{dc} = 0 \iff \Pr(\theta < c^* | x) = \Pr(\theta > c^* | x)$$

set the slope to zero: the masses balance

$$\implies c^* = \text{med}[\theta | x]$$

equal mass on both sides: the median

Why param-L1 beats param-MSE — amplitude: integrate the mean over the law

- **Param-MSE trains the amplitude toward $E[a | x]$** ; evaluate it on the spike-and-slab law, one manipulation per line:

$$E[a | x] = \int a p(a | x) da$$

the mean target, on the amplitude coordinate

$$\implies E[a | x] = \int a [\Pr(a=0 | x) \delta(a) + \Pr(a>0 | x) p(a | a>0, x)] da$$

insert the spike-and-slab law

$$\implies E[a | x] = \Pr(a=0 | x) \int a \delta(a) da$$

$$+ \Pr(a>0 | x) \int a p(a | a>0, x) da$$

linearity: pull the weights out

$$\implies \int a \delta(a) da = 0$$

sifting property of the point mass

$$\implies \int a p(a | a>0, x) da = E[a | a>0, x]$$

the slab integral is its own conditional mean

$$\implies E[a | x] = \Pr(a>0 | x) E[a | a>0, x]$$

evaluate

$$\implies c_{\text{MSE}}^* = \Pr(a>0 | x) E[a | a>0, x]$$

the occupied mean, dimmed by the posterior weight

$$\implies 0 < c_{\text{MSE}}^* < E[a | a>0, x]$$

both weights are strictly inside $(0, 1)$: a hedge

Why param-L1 beats param-MSE — amplitude: read the median off the CDF

- **Param-L1 trains the amplitude toward $\text{med}[a | x]$; read it off the spike-and-slab law, case by case:**

$\text{med}[a | x]$: any m with $\Pr(a \leq m | x) \geq \frac{1}{2}$ and $\Pr(a \geq m | x) \geq \frac{1}{2}$ the L1 target: the median's definition

$$\implies \Pr(a \leq m | x) = \Pr(a=0 | x)$$

$$+ \Pr(a > 0 | x) \Pr(a \leq m | a > 0, x)$$

the mixture's CDF: spike plus slab share

$$\implies \text{test } m=0 : \quad \Pr(a \leq 0 | x) = \Pr(a=0 | x), \quad \Pr(a \geq 0 | x) = 1$$

the CDF at 0: no slab mass there

$$\implies \Pr(a=0 | x) \geq \frac{1}{2} \implies \text{med}[a | x] = 0$$

both conditions hold: the empty truth

$$\implies \Pr(a=0 | x) < \frac{1}{2} : \text{the spike cannot reach } \frac{1}{2}$$

case split: the slab supplies the rest

$$\implies \Pr(a \leq \text{med} | a > 0, x) = \frac{1/2 - \Pr(a=0 | x)}{\Pr(a > 0 | x)}$$

set the CDF to $\frac{1}{2}$: the slab's share

$$\implies \text{med}[a | x] \text{ is a quantile of } p(a | a > 0, x)$$

an amplitude occupied slots actually take

$$\implies c_{\text{L1}}^* \in \{0\} \cup \text{supp } p(a | a > 0, x)$$

zero, or a real occupied amplitude

Why param-L1 beats param-MSE — position: the mean lands between the modes

- **The same two targets, now on the two-valued position law.** First the mean, one manipulation per line:

$$E[\mu | x] = \int \mu p(\mu | x) d\mu$$

the mean target, on the position coordinate

$$\implies E[\mu | x] = \int \mu [\Pr(\mu=\mu_1 | x) \delta(\mu-\mu_1) + \Pr(\mu=\mu_2 | x) \delta(\mu-\mu_2)] d\mu$$

insert the position law

$$\implies E[\mu | x] = \Pr(\mu=\mu_1 | x) \mu_1 + \Pr(\mu=\mu_2 | x) \mu_2$$

linearity, then sifting — as for the amplitude

$$\implies E[\mu | x] - \mu_1 = (\Pr(\mu=\mu_1 | x) - 1) \mu_1 + \Pr(\mu=\mu_2 | x) \mu_2$$

subtract μ_1 from both sides

$$\implies \Pr(\mu=\mu_1 | x) - 1 = -\Pr(\mu=\mu_2 | x)$$

the two masses sum to one

$$\implies E[\mu | x] - \mu_1 = \Pr(\mu=\mu_2 | x) (\mu_2 - \mu_1)$$

collect

$$\implies \mu_2 - E[\mu | x] = \Pr(\mu=\mu_1 | x) (\mu_2 - \mu_1)$$

the same three lines, started from μ_2

$$\implies \text{both gaps} > 0 \implies \mu_1 < E[\mu | x] < \mu_2$$

the mean is strictly between the modes

$$\implies \min_j |E[\mu | x] - \mu_j| = \min_j \Pr(\mu=\mu_j | x) \cdot (\mu_2 - \mu_1)$$

the nearer mode carries the smaller mass

Why param-L1 beats param-MSE — the tail's share of the gradient

Define the tail's share of the batch pull, then evaluate it population by population (tail = the εN pinned coordinates, core = the $(1-\varepsilon)N$ learning ones):

$$S = \frac{\sum_{i \in \text{tail}} |\partial \ell / \partial c_i|}{\sum_{i \in \text{tail}} |\partial \ell / \partial c_i| + \sum_{i \in \text{core}} |\partial \ell / \partial c_i|}$$

- **MSE — sum the slopes.**

tail: εN terms, each $2|r_i| \approx 2R$

$$\Rightarrow \sum_{\text{tail}} = \varepsilon N \cdot 2R$$

core: $(1-\varepsilon)N$ terms, each $2|r_i| \approx 2\sigma_r$

$$\Rightarrow \sum_{\text{core}} = (1-\varepsilon)N \cdot 2\sigma_r$$

$$\Rightarrow S_{\text{MSE}} = \frac{\varepsilon N \cdot 2R}{\varepsilon N \cdot 2R + (1-\varepsilon)N \cdot 2\sigma_r}$$

$$\Rightarrow S_{\text{MSE}} = \frac{\varepsilon R}{\varepsilon R + (1-\varepsilon)\sigma_r} \quad (\text{cancel } 2N)$$

- **L1 — the same sums, then the limit.**

every term: $|\frac{\partial \ell}{\partial c_i}| = 1$

$$\Rightarrow \sum_{\text{tail}} = \varepsilon N, \quad \sum_{\text{core}} = (1-\varepsilon)N$$

$$\Rightarrow S_{\text{L1}} = \frac{\varepsilon N}{\varepsilon N + (1-\varepsilon)N} = \frac{\varepsilon N}{N} = \varepsilon$$

training succeeds: $\sigma_r \rightarrow 0$

$$\Rightarrow (1-\varepsilon)\sigma_r \rightarrow 0$$

$$\Rightarrow S_{\text{MSE}} \rightarrow \frac{\varepsilon R}{\varepsilon R} = 1, \quad S_{\text{L1}} = \varepsilon \text{ always}$$

Why param-L1 beats param-MSE — the checks: Huber as a gap probe, and the gate

- **Huber interpolates the two losses:** quadratic inside δ , linear outside — configured here at $\delta = 0.5$ (normalised units). If the tail story is right, every cell must order $\text{MSE} \leq \text{Huber} \leq \text{L1}$, and Huber's share of the gap, $\frac{\text{Huber}-\text{MSE}}{\text{L1}-\text{MSE}}$, reads out where each cell's tail gaps sit against δ :

grid cell	p-MSE	p-Huber	p-L1	Huber's share
ResUNet conv·sorted	0.937	0.961	0.982	53 %
ResUNet conv·Hungarian	0.924	0.961	0.982	64 %
ResUNet set-pred·Hungarian	0.939	0.972	0.988	67 %
Att-UNet conv·sorted	0.855	0.853	0.936	−2 %
Att-UNet conv·Hungarian	0.849	0.849	0.921	0 %
Att-UNet set-pred·Hungarian	0.892	0.913	0.942	42 %
DeepLabV3+ conv·sorted	0.856	0.971	0.983	91 %
DeepLabV3+ set-pred·Hungarian	0.934	0.984	0.992	86 %

bold = best in row, marked only where best and worst differ by more than 5%. The ordering holds in seven of eight cells (the exception is a 0.002 gap). The share column splits by regime: where the tail gaps exceed δ , Huber is L1-like and recovers most of the gap (53–91%); both ungated Att-UNet rows recover none — Huber behaved as pure MSE, which the tail story can only read as tail gaps sitting *inside* $\delta = 0.5$. That is a prediction the residual histograms must confirm, not a fit. The gate is the second check: it cut the amplitude hedge architecturally (the amplitude frames), so the gated rows must lose the hedge's backbone-dependent part — and they do: L1–MSE collapses to a uniform 0.049–0.058 band on the three set-pred rows, while the ungated rows scatter from 0.045 to 0.127. The surviving band is the position-and-matcher tail no gate can remove.

Why param-L1 beats param-MSE — what the checks cannot yet say

The ordering is established; the mechanism split is not. What the table measures, and the runs that close the case:

- **What the table does and does not measure.**

- the score is the comparison's min-max composite: shared anchors, rank units — *orderings* across cells are meaningful, gap sizes are not physical units;
- one seed per cell — the 0.002 violation has no seed spread behind it; the real evidence is the sign test: the ordering holds in seven of eight independent cells;
- the optimum half prices a *fixed* label law; the Hungarian matcher re-derives the target from the prediction each step — the hedge pricing is exact for sorted, an approximation for Hungarian. The Hungarian gaps are not smaller, so the gradient half must carry weight there.

- **The runs that close it.**

- matched-residual histograms per objective (the loss-pair sweep emits them): σ_r , R , ε become measurements instead of assumed scales; excess kurtosis is the direct tail read; where the tail mass sits against $\delta = 0.5$ adjudicates the Att-UNet rows;
- the sweep's weight axis: param-MSE at larger w lowers its noise floor ($2w|r| = \eta$ parks at $|r| = \eta/2w$) but moves no wrong target — if a retuned weight closes the gap, the floor and fade carried the result; if the gap survives every w , the hedge did.

Two mechanisms drive one ordering — hedge at the optimum, floor-and-fade in the gradient; the sweep's histograms and weight axis are what separates them.